

SABA A. FARAHANI

AI Researcher | LLM Agents • Knowledge Graphs • Multimodal Learning • Causal AI
fazizaba@uci.edu • LinkedIn • GitHub • Google Scholar • sabafarahani.com

EDUCATION

PhD, Electrical Engineering and Computer Science • University of California, Irvine	2023–Present
Research: Trustworthy Personalized Agents: From Multimodal Data to Verified Action	
MS, Electrical Engineering and Computer Science • Sharif University of Technology	2020–2022
BS, Electrical Engineering • University of Tehran	2015–2019

RESEARCH AND INDUSTRY EXPERIENCE

UC Irvine Institute for Future Health	2025–Present
Graduate Researcher Research Intern, Summer 2026	
- Designed and implemented a knowledge-guided LLM agent that identifies missing patient context and generates targeted follow-up questions before downstream response generation. - Designed and implemented T2D-Bench, using a clinical-lifestyle knowledge graph and deterministic evidence checks to verify and revise LLM responses against explicit clinical evidence requirements; evaluated on 100 structured cases. - Developing Personal Care Utility (PCU), a multi-agent system combining wearable signals, clinical context, and event graphs for personalized type 2 diabetes care. - Designed and implemented an event-graph mining agent integrating wearable and physiological signals with clinical reports to detect health events, identify event relationships, estimate next-event probabilities, and interpret clinical significance.	
UC Irvine Sue & Bill Gross School of Nursing	2025
Graduate Researcher • Clinical Trial Matching	
Designed and implemented a multi-LLM pipeline combining retrieval-augmented reasoning and conversational AI to match patient profiles against clinical-trial eligibility criteria.	
NaviGait Company	2025
AI and Robotics Research Intern	
Built data-collection and validation pipelines for care-companion robotics; deployed camera and health sensors for dementia monitoring with UCI Nursing collaborators.	
Hyundai Motor Company	2023–2024
Research Collaboration and Internship • Autonomous Systems	
- Collected and processed sensor data; designed and implemented predictive models for LiDAR contamination and investigated sensor-cleaning systems. Evaluated the models with the Hyundai team. - Analyzed LiDAR, camera, GPS, and weather data; implemented RGB-infrared image fusion to investigate object detection under low-visibility conditions.	

INNOVATION AND AWARDS

- Co-inventor, UCI invention disclosure (2026)** — Causal AI for personalized health. Developed an interpretable framework for multimodal health assessment, evaluated across two cohorts totaling **2,365 participants**.
- Best Paper Award**, ASME InterPACK 2024 — Predictive modeling for LiDAR sensor reliability.

TECHNICAL SKILLS

Programming and Tools	Python, C/C++, PyTorch, TensorFlow, OpenCV, Linux
AI and machine learning	Large language models (LLMs), multi-agent systems, retrieval-augmented reasoning, knowledge graphs, causal discovery, multimodal learning, biomedical signal processing, time-series analysis